



Higher Diploma in Computer Science

Computer Systems & Networks



Working with Files

NDG Linux Essentials - Lab 8: Managing Files and Directories

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Learning Outcomes:

- This class will cover the basics of the file management such as:
 - Globbing
 - Copying files
 - Moving files
 - Renaming files
 - Removing files

Globbering



Globbing

- *Glob characters* are often referred to as *wild cards*.
- These are symbol characters that have special meaning to the shell.
- Globs are powerful because they allow you to specify patterns that match filenames in a directory:
 - Instead of manipulating a single file at a time, you can easily execute commands that will affect many files.

Globber - The Asterisk *

- The asterisk character is used to represent zero or more of any character in a filename.

```
compsys@compsys-virtualbox:~$ ls -l
total 52
drwxrwxr-x 2 compsys compsys 4096 Aug 31 2022 Desktop
drwxr-xr-x 2 compsys compsys 4096 Aug 25 22:32 Documents
drwxr-xr-x 2 compsys compsys 4096 Aug 25 22:32 Downloads
drwxr-xr-x 2 compsys compsys 4096 Aug 25 22:32 Music
drwxr-xr-x 2 compsys compsys 4096 Aug 25 22:32 Pictures
drwxr-xr-x 2 compsys compsys 4096 Aug 25 22:32 Public
drwxrwxr-x 2 compsys compsys 4096 Aug 30 12:33 Sept
drwx----- 3 compsys compsys 4096 Aug 25 21:34 snap
-rwxr-x--x 1 compsys compsys 248 Aug 29 13:54 stats
-rwxr-xr-x 1 compsys compsys 29 Aug 31 10:55 tempdate
drwxr-xr-x 2 compsys compsys 4096 Aug 25 22:32 Templates
drwxrwxr-x 3 compsys compsys 4096 Aug 31 13:01 tmp
drwxr-xr-x 2 compsys compsys 4096 Aug 25 22:32 Videos
compsys@compsys-virtualbox:~$ touch Semester1 Semester1Results Semester2 Semester3 SemanticDetails
compsys@compsys-virtualbox:~$ ls -l Sem*
-rw-rw-r-- 1 compsys compsys 0 Aug 31 16:00 SemanticDetails
-rw-rw-r-- 1 compsys compsys 0 Aug 31 16:00 Semester1
-rw-rw-r-- 1 compsys compsys 0 Aug 31 16:00 Semester1Results
-rw-rw-r-- 1 compsys compsys 0 Aug 31 16:00 Semester2
-rw-rw-r-- 1 compsys compsys 0 Aug 31 16:00 Semester3
compsys@compsys-virtualbox:~$
```

Globbing - The Question Mark ?

- The question mark character matches exactly one character, no more and no less.
- The asterisk and question mark could also be used together to look for files

```
compsys@compsys-virtualbox:~$ ls -l Sem?????  
-rw-rw-r-- 1 compsys compsys 0 Aug 31 16:00 Semester1  
-rw-rw-r-- 1 compsys compsys 0 Aug 31 16:00 Semester2  
-rw-rw-r-- 1 compsys compsys 0 Aug 31 16:00 Semester3  
compsys@compsys-virtualbox:~$
```

Globbing - Brackets []

- Brackets are used to match a single character by representing a range of characters that are possible match characters.

```
compsys@compsys-virtualbox:~$ ls [SD]*
SemanticDetails Semester1 Semester1Results Semester2 Semester3

Desktop:
computer.desktop network.desktop qterminal.desktop trash-can.desktop user-home.desktop

Documents:

Downloads:

Sept:
compsys@compsys-virtualbox:~$
```

- Brackets can also be used to represent a range of characters by using the - character

```
compsys@compsys-virtualbox:~$ ls [S-W]*
```


Globbering - Listing With Globbs

- Use the `-d` option, which tells the `ls` command to display the name of directories, instead of their contents:

```
compsys@compsys-virtualbox:~$ ls -d [SD]*  
Desktop Documents Downloads SemanticDetails Semester1 Semester1Results Semester2 Semester3 Sept  
compsys@compsys-virtualbox:~$
```

Globbering - Exclamation Point !

- The exclamation point is used in conjunction with the square brackets to negate a range.

```
compsys@compsys-virtualbox:~$ ls [!S]*
stats  tempdate

Desktop:
computer.desktop  network.desktop  qterminal.desktop  trash-can.desktop  user-home.desktop

Documents:

Downloads:

Music:

Pictures:

Public:

snap:
firefox

Templates:

tmp:
tutorial

Videos:
compsys@compsys-virtualbox:~$
```

Copying Files



Copying Files

- The `cp` command is used to copy files. It requires a source and a destination.

```
cp [source] [destination]
```

- `cp` will **overwrite** any existing files by default

```
compsys@compsys-virtualbox:~$ cp Semester1Resu* Sept
compsys@compsys-virtualbox:~$ ls Sept
Semester1Results
compsys@compsys-virtualbox:~$ █
```

Copying Files – v (Verbose Mode)

- The `-v` option will cause the `cp` command to produce output if successful.
- An example of the `-v` option used with the `cp` command:
- When the destination is a directory, the resulting new file will have the same name as the original file.
- If you want the new file to have a different name, you must provide the new name as part of the destination.

```
compsys@compsys-virtualbox:~$ cp -v Semester2 Sept  
'Semester2' -> 'Sept/Semester2'
```

Copying Files - Avoid Overwriting Data

- The `cp` command can be destructive to existing data if the destination file already exists.

```
compsys@compsys-virtualbox:~$ cp -vi Semester2 Sept  
cp: overwrite 'Sept/Semester2'? █
```

- With the `-i` (interactive) option, the `cp` will prompt before overwriting a file (`y` (yes) or `n` (no))

Copying Directories

- Using the `cp` command to copy directories will result in an error message:

```
sysadmin@localhost:~$ cp -n /etc/skel/* ~  
cp: omitting directory `/etc/skel/.'  
cp: omitting directory `/etc/skel/..'
```

- However, the `-r` (recursive) option to the `cp` command will have it copy both files and directories.

Be careful

The entire directory structure will be copied.

This could result in copying a lot of files and directories!

Moving/ Renaming Files



Moving Files: the `mv` command

- The syntax for the `mv` command is much like the `cp` command:

```
mv [source] [destination]
```

- The `mv` command also renames a file.
 - The name of the file will change only if a destination file name is also specified.

```
compsys@compsys-virtualbox:~$ mv Semester1 Sept/ClassList
compsys@compsys-virtualbox:~$ ls Sept
ClassList Semester1Results Semester2
compsys@compsys-virtualbox:~$
```

Moving Files - Additional mv Options

- Like the `cp` command, the `mv` command provides the following options:

Option	Meaning
<code>-i</code>	Interactive: Ask if a file is to be overwritten.
<code>-n</code>	Do not overwrite a destination files' contents.
<code>-v</code>	Verbose: Show the resulting move.

Removing Files



Deleting Files

- To delete a file, use the `rm` command
- Be careful if using glob/wildcard characters

```
compsys@compsys-virtualbox:~$ ls
Desktop  Downloads  Pictures  SemanticDetails  Semester2  Sept  stats  Templates  Videos
Documents Music      Public   Semester1Results Semester3  snap  tempdate tmp
compsys@compsys-virtualbox:~$ rm Sem*
compsys@compsys-virtualbox:~$ ls
Desktop  Downloads  Pictures  Sept  stats  Templates  Videos
Documents Music      Public   snap  tempdate tmp
compsys@compsys-virtualbox:~$
```

Deleting Directories

- The `rm` command can be used to delete directories. However, the default usage (no options) of the `rm` command will fail to delete a directory:

```
compsys@compsys-virtualbox:~$ ls tmp
tutorial
compsys@compsys-virtualbox:~$ rm -i tmp
rm: cannot remove 'tmp': Is a directory
compsys@compsys-virtualbox:~$ rm -ri tmp
rm: descend into directory 'tmp'? y
rm: descend into directory 'tmp/tutorial'? y
rm: remove directory 'tmp/tutorial/Oct'? y
rm: remove directory 'tmp/tutorial/Nov'? n
rm: descend into directory 'tmp/tutorial/Dec'? y
rm: descend into directory 'tmp/tutorial/Dec/Part1'? y
rm: descend into directory 'tmp/tutorial/Dec/Part1/Outputs'? y
rm: remove regular empty file 'tmp/tutorial/Dec/Part1/Outputs/Part1.txt'? y
rm: remove regular file 'tmp/tutorial/Dec/Part1/Outputs/output.txt'? █
```

- To delete a directory, use the `-r` (recursive) option to the `rm` command:
Important: recall the `-i` option, why?